

00 — Overview and Reading Map

What you'll learn: what a Gaussian Process is in one paragraph, why it suits underwater positioning, and how the rest of these docs fit together.

Prereqs: none.

Why it matters here: orientation before the detail.

A Gaussian Process in one paragraph

A **Gaussian Process (GP)** is a probability distribution over functions. Instead of fitting a single curve $f(x)$, a GP says: "any finite set of function values $f(x_1), \dots, f(x_n)$ is jointly Gaussian," fully specified by a **mean function** $m(x)$ (often 0) and a **covariance/kernel function** $k(x, x')$ that says how correlated two outputs are based on how close their inputs are. Given some noisy observations, conditioning this joint Gaussian on the data yields a **posterior**: at any new input x^* you get a predicted **mean** and a predicted **variance**. The variance is small near data and grows as you move away — built-in, honest uncertainty. That uncertainty is the property that makes GPs valuable for sensor fusion.

Why this is the right tool underwater

Underwater navigation has three painful facts:

1. **No GPS.** Position is obtained by integrating velocity/acceleration → unbounded drift.
2. **The DVL drops out and lies.** The Doppler Velocity Log is the workhorse velocity sensor, but it

loses bottom-lock (soft mud, steep slopes, altitude limits, bubbles) and emits outliers.

3. **Everything is uncertain and asynchronous.** IMU at ~100 Hz, DVL at ~1 Hz, magnetometer

sporadically; each with its own noise.

A GP addresses all three: it produces a **continuous, uncertainty-aware** estimate of velocity (or

position) over time, so it can **denoise** good DVL, **reject** outliers, and **bridge** gaps with a variance that grows — telling the downstream Kalman filter exactly how much to trust the bridged

value. And through the SPDE duality (doc 06) it can be taught **physics** and **boundary conditions**;

through the Kalman duality (doc 05) it can run **cheaply on the vehicle** (doc 08).

The three themes the user asked about

- **GPs for AUV navigation** — the practical "how does it plug in and help" — doc 04, plus all of implementation/.

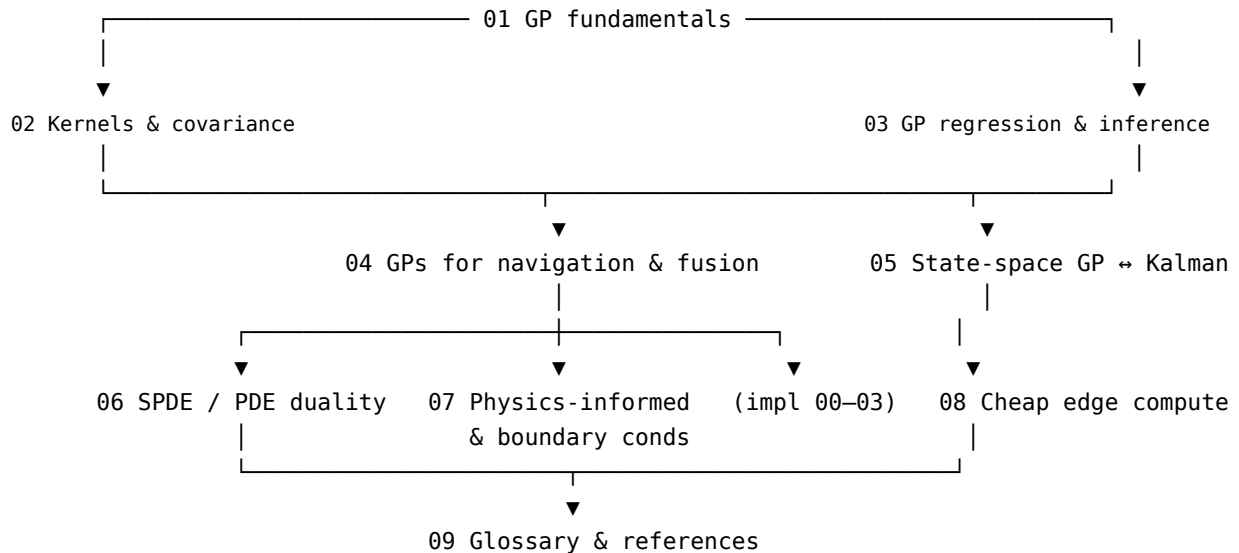
- **The second-order-PDE / SPDE duality** — "incorporate physical processes and boundary conditions" — docs 06 and 07. This is the deep idea: a Matérn-covariance GP *is the solution of a

stochastic PDE*, so changing the PDE changes the physics the GP assumes.

- **Cheap edge compute** — docs 05 and 08. The key trick: a temporal GP with the right kernel is

mathematically identical to a Kalman filter, which is $O(n)$ and tiny.

Dependency-ordered reading map



- **Just want the build?** Read 01 → 03 → 04, then implementation/.
- **Care most about the PDE duality?** Read 01 → 02 → 06 → 07.
- **Care most about edge compute?** Read 01 → 02 → 05 → 08.

How the math is written here

Equations are plain text/Markdown with backticks, e.g. $k(x, x') = \sigma^2 \exp(-(x-x')^2/(2\tau^2))$.

Vectors

are bold in prose ("**x**"), matrices are capitals (**K**). Where a symbol matters, it is defined inline and collected in doc 09's glossary.

Key takeaways

- A GP = distribution over functions = mean + kernel; its output is a **mean and a variance**.
- The variance (uncertainty) is why it helps sensor fusion: it tells the Kalman filter how much to

trust each value, including bridged values during DVL dropouts.

- Two dualities unlock the advanced asks: **GP ↔ Kalman** (cheap edge) and **GP ↔ SPDE** (physics + boundary conditions).

Further reading

- Rasmussen & Williams, Gaussian Processes for Machine Learning (free online) — the canonical text.
- Continue to 01 — GP fundamentals.